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Acknowledgements

I would like to thank the author of this template, who has saved me a lot of work (...you don't actually have to)

Abstract

My abstract, which will be 300 words or fewer.

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List of abbreviations

CAO	cold-air outbreak
HALO-(AC)^3	Project on Arctic Amplification Climate Relevant Atmospheric and Surface Processes and Feedback Mechanisms
lat.	latitude

List of symbols

g	Graupel (as a model hydrometeor species).
i	Ice crystals (as a model hydrometeor species). Note this refers to the species only, and not to the total frozer water content, which would include snow and graupel.
r	Rain (as a model hydrometeor species).
s	Snow (as a model hydrometeor species).
w	Cloud water (as a model hydrometeor species). Note this refers to the species only, and not to the total liquid water content, which would include rain.
Z	Universal coordinated time.
$\overline{a}, \overline{b}, \ldots$	Arbitrary constants used in parametrisations of average values. Values are defined at source with respect to their parametrisations.
$a, b, \ldots$	Arbitrary constants used in parametrisations. Values are defined at source with respect to their parametrisations.
A_{d}	Surface area of filter in contact with a droplet in a droplet-freezing assay.
A_{f}	Surface area of filter exposed to aerosol in a droplet-freezing assay.
N_{INP}	Number concentration of ice-nucleating particles.
$N_{\mathrm{INP,\ con.}}$	Number concentration of contact-mode ice-nucleating particles.
$N_{\mathrm{INP,\ imm.}}$	Number concentration of immersion-mode ice-nucleating particles.
x	Placeholder variable for an unspecified CASIM hydrometeor or aerosol tracer.
y	Placeholder variable for an unspecified CASIM hydrometeor or aerosol tracer.
$\mathcal{N}_x$	Particle size distribution of hydrometeor species x.
η_{c}	Efficiency of contact freezing (model parameter).
η_{i}	Efficiency of immersion freezing (model parameter).

Chapter 1

Introduction

1.1 Thesis to-do

Keep going to see some actual stuff. Note that, like for previous chapter-level entries, there is a blank page after this to ensure that your chapter starts on an odd-numbered page, cause that's a convention for some reason.

Chapter 2

High ice-nucleating particle concentrations associated with Arctic haze in springtime cold-air outbreaks

Abstract

A section for the abstract that deliberately does not appear in the ToC. Work in this excerpt was taken from Raif, E. N., Barr, S. L., Tarn, M. D., McQuaid, J. B., Daily, M. I., Abel, S. J., Barrett, P. A., Bower, K. N., Field, P. R., Carslaw, K. S., and Murray, B. J.: High ice-nucleating particle concentrations associated with Arctic haze in springtime cold-air outbreaks, Atmos. Chem. Phys., 24, 14045–14072, https://doi.org/10.5194/acp-24-14045-2024, 2024.

2.1 Excerpt from paper

Figure 2.1a shows the potential sources of INPs for CAO clouds in the Northern Hemisphere. Potential Arctic sources of INPs are biogenic material from sea-spray (Wilson et al., 2015; DeMott et al., 2016; McCluskey et al., 2018) and terrestrial INPs from high-latitude sources (Kawai et al., 2023). Potential terrestrial sources of INPs include glacial outwash sediments (Tobo et al., 2019; Xi et al., 2022; Barr et al., 2023), sandy deserts in Iceland (Sanchez-Marroquin et al., 2020), boreal forests (Brasseur et al., 2022), Arctic surface vegetation (Pereira Freitas et al., 2023) and thawing permafrost (Creamean et al., 2020). In addition, Arctic aerosol can have lifetimes of many weeks, which results in the build up of Arctic haze in winter and spring (Stohl, 2006). Accordingly, it is also possible that INPs entering CAOs may have been transported into the Arctic from lower latitudes where they reside in the stable Arctic atmosphere before being transported out again in CAOs (Carslaw, 2022). Different types of INPs have different characteristic ranges of freezing temperatures (Kanji et al., 2017). This means understanding the aerosol sources that contribute to the INP population in CAO clouds can reduce the uncertainty in the cloud-phase feedback.

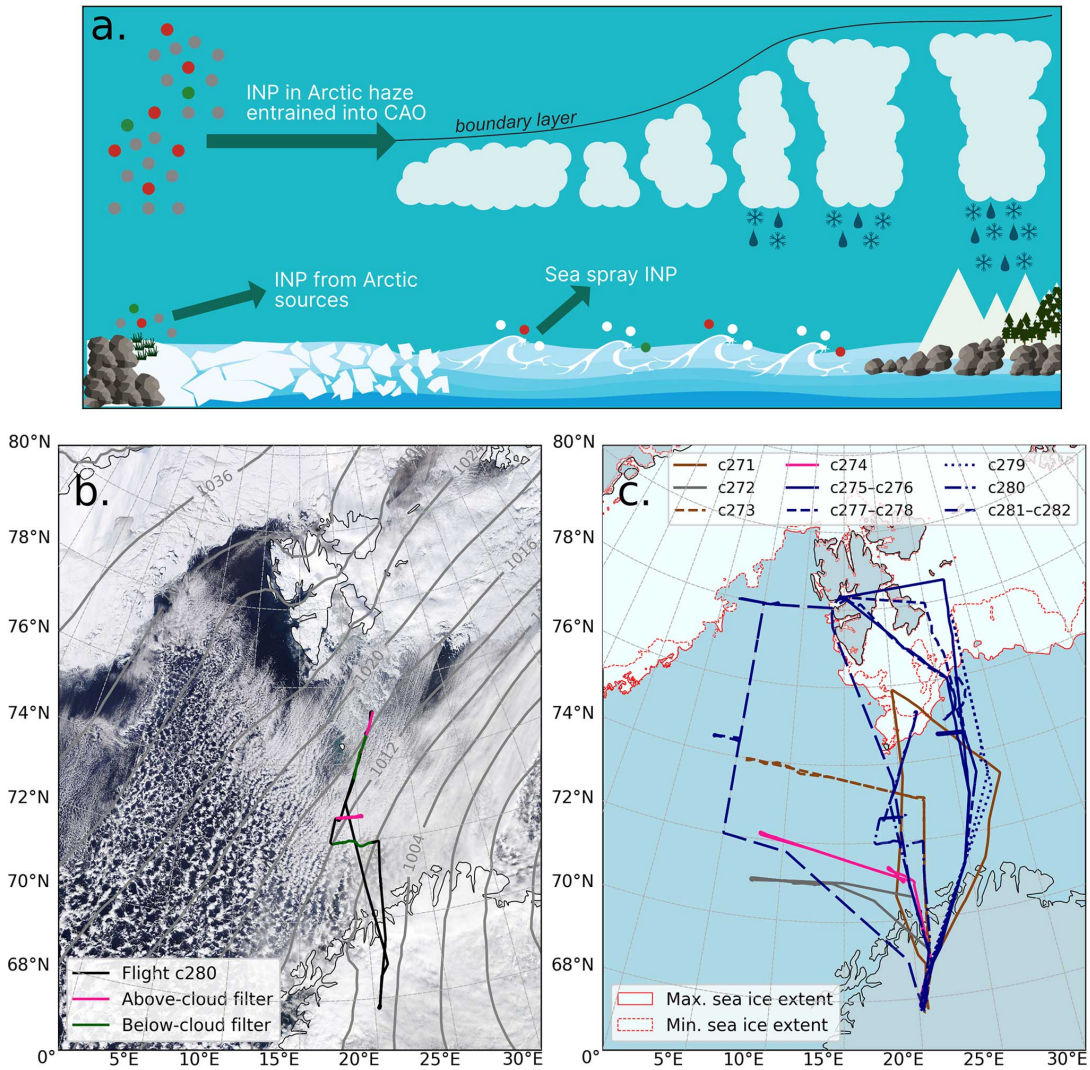

Figure 2.1: For brevity, the full caption is excluded but thanks to MODIS, the US National Ice Center (MASIE) and ECMWF (ERA5) and the Facility for Airborne Atmospheric Measurements for data used in these plots.

Aerosol concentrations near the sea surface ($<$ 500 m a.s.l.) were consistently 100 cm^{-3} to 200 cm^{-3}.

2.1.1 Results

These sampling locations were considered to represent air below, above and before the development of the atmospheric boundary layer over sea respectively. The locations, altitudes and purpose of each filter measurement are shown in Table 2.1.

$$\overline{N}_{\mathrm{INP}} = \overline{a} \exp\left[\overline{b}\,(\overline{c} - T)^{\overline{d}}\right], \tag{2.1}$$

Using the values of $N_{\mathrm{INP}}(-15\,°\mathrm{C})$ and the total aerosol surface area S_{aer}, we calculated the factors of change in INP concentration (F_{INP}), ice-nucleating activity (F_{S}) and surface

Table 2.1: Times, locations and sampling intent of all filter samples made during the campaign.

Sample ID	Run type	Date (mm-dd)	Start (UTC)	End (UTC)	Mean alt. m	Mean lat. (° N)	Mean lon.(° E)	CAO flow	Notes
c271r1	above cloud	03-11	09:40:07	10:07:24	1747	74.3	20.3	W	
c271r2	below cloud	03-11	10:18:03	10:42:48	39	75.5	21.5	W	2 min pause during aircraft turning
c271r3	below cloud	03-11	11:18:05	11:38:38	49	72.7	27.9	W	filter processed 3 d after collection two pauses for precipitation
c272r6	above cloud	03-16	12:07:25	12:34:15	2957	71.0	10.5	N/A	warm-air intrusion conditions
c273r1	above cloud	03-19	12:01:25	12:38:51	3403	73.8	11.2	W	
c273r2	below cloud	03-19	12:54:07	13:13:35	415	73.9	8.7	W	two short pauses to avoid showers
c274r1	above cloud	03-21	13:47:10	14:07:23	2672	71.8	12.6	NW	
c274r3	below cloud	03-21	15:18:08	15:41:22	165	71.2	19.3	NW	two short pauses for aircraft turning and one pause to avoid showers
c275r1	below cloud	03-23	10:03:37	10:28:22	162	76.5	23.8	N	
c276r2	upstream	03-23	12:55:29	13:20:29	1711	78.5	22.0	N	
c276r4	below cloud	03-23	14:32:27	14:55:27	156	74.6	25.2	N	
c277r1	below cloud	03-24	10:30:52	10:59:06	159	76.7	22.8	N	two pauses for altitude increases cloud in very early stages of formation
c278r2	upstream	03-24	13:02:05	13:26:16	1725	78.2	21.3	N	
c278r4	below cloud	03-24	14:07:00	14:31:56	333	75.1	26.0	N	2 min pause for turning due to precip.
c279r1	below cloud	03-25	11:23:21	11:44:12	155	76.9	28.0	N	
c279r2	above cloud	03-25	11:48:26	12:08:44	1323	75.4	28.0	N	
c280r1	above cloud	03-29	09:29:00	09:49:17	1573	74.8	21.6	N	
c280r2	below cloud	03-29	09:59:55	10:15:58	327	74.1	20.6	N	
c280r3	above cloud	03-29	10:34:01	10:54:06	2013	72.6	19.4	N	2 min pause during aircraft turning
c280r4	below cloud	03-29	11:05:35	11:24:51	334	72.0	19.3	N	1 min pause to avoid showers
c282r1	above cloud	03-30	12:08:32	12:28:27	1761	78.0	7.5	N	
c282r3	below cloud	03-30	14:02:12	14:26:08	307	74.4	3.5	N	4 min pause during aircraft turning

Flight	Upwind filter	Downwind filter	$\Delta\varphi$	F_{INP}	F_{S}	F_{aer}
c275–276	c275r1, Sc	c276r4t, Sc	1.92	0.40	0.52	0.75
c280	c280r2, Sc	c280r4t, precipitating Cu	2.17	0.35	0.19	1.85
c281-282	c282r2, Sc	c282r3t, precipitating Sc–Cu	3.58	0.68	0.39	1.69

Table 2.2: Factors of change in the INP concentration at −15 °C (F_{INP}), INP concentration normalised by surface area at −15 °C (F_{S}) and total aerosol surface area (F_{aer}) between three pairs of below-cloud filter measurements.

area (F_{aer}) between the upwind and downwind measurements:

$$F_{\mathrm{INP}} = \frac{N_{\mathrm{INP}}\,(-15\,°\mathrm{C},\ \mathrm{downwind})}{N_{\mathrm{INP}}\,(-15\,°\mathrm{C},\ \mathrm{upwind})}, \tag{2.2}$$

$$F_{\mathrm{S}} = \frac{N_{\mathrm{INP}}\,(-15\,°\mathrm{C},\ \mathrm{downwind})/S_{\mathrm{aer}}\,(\mathrm{downwind})}{N_{\mathrm{INP}}\,(-15\,°\mathrm{C},\ \mathrm{upwind})/S_{\mathrm{aer}}\,(\mathrm{upwind})}, \tag{2.3}$$

$$F_{\mathrm{aer}} = \frac{S_{\mathrm{aer}}\,(\mathrm{downwind})}{S_{\mathrm{aer}}\,(\mathrm{upwind})}. \tag{2.4}$$

These values are displayed in Section 2.1.1 with the difference in latitude between the measurements, $\Delta\varphi$. Both the INP concentration and the ice-nucleating activity decreased with latitude. The total surface area of the aerosol increased in two flights by 69 % and 85 % but decreased by 25 % in one pair. The upwind filters were sampled under stratocumulus conditions, while the downwind filters were in a mixture of stratocumulus and cumulus, with precipitation in two cases (c280r4 and c282r3).

Chapter 3

My second proper chapter

3.1 An annoyingly long table

Sometimes, you need a particularly annoyingly long table. Like this one which is in my thesis.

Table 3.1: A summary of the microphysical processes included in CASIM. Condensation and evaporation of cloud water is controlled outside CASIM by the bi-modal cloud fraction scheme of vWB21.

Process	Description	Hydrometeor
Accretion	Collection of a (usually) smaller hydrometeor by a (usually) larger hydrometeor.	cloud water by all others; ice by all but cloud water; rain by snow and graupel; snow by graupel
Activation	Activation of a cloud condensation nucleus into a droplet. By default, this is performed according to the ARG00 scheme.	cloud water only
Aggregation	Self-collection of precipitation-size hydrometeors of a single species.	rain and snow
Autoconversion	Self-collection of a smaller species into the larger species of the same phase.	cloud water to rain; ice to snow
Deposition	Growth of ice particles by vapour deposition when air is supersaturated with respect to ice. Deposition ice nucleation is not included in this.	all frozen species
Evaporation	Evaporation rate of raindrops when entering air subsaturated with respect to water.	rain only

Continued on next page...

Table 3.1 – continued from previous page

Process	Description	Hydrometeor
Ice nucleation (homogeneous)	Freezing of liquid droplets at temperatures below −38 °C.	cloud water to ice; rain to graupel
Ice nucleation (heterogeneous)	Fully described elsewhere. By default, DM15 is used when aerosol is processed and C86 if not. Additionally, B53 is used for the heterogenous freezing of raindrops, though this is contained within the homogeneous nucleation subroutine.	cloud water to ice
Melting	Melting of frozen species to become rain. Instantaneous for ice, but not for snow and graupel, which adhere to a parameterised melting rate.	ice to rain; snow to rain; graupel to rain
Secondary ice production	Increase in ice crystal concentration through the Hallett-Mossop process between −2.5 °C and −7.5 °C. The production rate is controlled by snow and graupel production rates.	all frozen species
Sedimentation	Species falling into lower model levels under gravity.	all species
Sublimation	Sublimation of frozen hydrometeors when entering air subsaturated with respect to ice.	all frozen species

Chapter 4

My third proper chapter

Placeholder

4.1 A section

A section

Appendix A

INP spectra and associated aerosol size distributions for each aerosol sample during ACAO

A.1 INP spectra and aerosol size distributions

Some content.

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